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REVIEW

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When is a sample a urine sample? Markers for urine sample authenticity assessment in sports drug testing

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ABSTRACT

Human routine doping controls rely to a considerable extent on urine samples, and test methods have been optimized for decades to allow for comprehensive analyses with adequate analytical sensitivities. However, attempts of sample manipulation, presumably in order to escape an adverse analytical finding, were reported to include adulteration as well as substitution, which has revived the debate as to when a sample submitted as urine specimen for sports drug testing purposes is, indeed, to be considered as 'urine.' In consideration of selected case reports as well as analytical parameters routinely acquired with doping control urine samples, options concerning guidance in that context are discussed, as detecting and reporting urine substitution and urine adulteration require laboratory approaches that might differ considerably from protocols applied for, e.g. the confirmation of the presence of a prohibited substance. In particular, commonly monitored markers characterizing urinary specifics such as pH, specific gravity, and the presence of a human urinary steroid profile and 5-aminoimidazole-4-carboxamide ribonucleoside (AICAR) in established reference ranges could assist in defining acceptance criteria for urine samples and, thereby, guidelines when a specimen does not qualify as human urine.

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Doping; sport; manipulation; biomarker; steroid profile

1. Introduction

Urine samples represent the majority of specimens that are routinely submitted to doping controls, complemented by blood (i.e., whole blood or serum) as well as dried blood spot (DBS) samples (Figure 1a) [1,2]. History taught, repeatedly, that attempts to undermine doping control analytical methods have been made, either by the manipulation of an athlete's urine sample (e.g., by the addition of ethanol [3] or proteases [4]) or by sample substitution [5–7], but the inherent issue remains that effectively conducted sample manipulation might be uncovered only months or years after the fact [8] (if at all), which is why annual statistics [1,2] do not accurately reflect the numbers of anti-doping rule violations attributable to, e.g., urine sample substitution. A representative example on investigations into suspected sample manipulation, which is not accounted for in annual laboratory statistics, was documented under the World Anti-Doping Agency's (WADA's) "Operation Arrow" [8], where a total of 53 samples (related to 39 athletes) was included in a follow-up review. Eleven samples were eventually proven to be substituted, 15 were "very likely" substituted but already discarded at the time of the investigation, and 27 were found to be not substituted (Figure 1b).

At the time of the sample collection, strict regulations as to the overall procedure and suitability of a urine sample for doping control purposes apply, including the athlete's obligation to wash hands, an unobstructed view of the sample leaving the athlete's body, the permanent observation of the sample after provision until the sealing of the sample is completed, a minimum sample volume of 90 mL, and a suitable specific gravity. For the latter, when determined with a refractometer, the measured value must be 1.005 or higher at a total volume of less than 150 mL of urine; at a volume of 150 mL and above, the specific gravity must be at least 1.003 [9].

Drug testing program guidelines for various purposes such as workplace safety, federal employment examination, forensic investigations, or clinical diagnostics recommend the assessment of standard markers including temperature, creatinine, and pH in support of specimen validity tests (SVTs) [10,11]. More recently, additional complementary target analytes characteristic for human urine as well as those for surrogates (i.e., synthetic urine as well as other liquids submitted as urine substitute) and adulterants were proposed, facilitating the identification of manipulated specimens [12–14]. In doping controls, such complementary criteria have been indicated in the Technical Document on the "Measurement and Reporting of Endogenous Anabolic Androgenic Steroid (EAAS) Markers of the Urinary Steroid Profile" [15] with for example specific gravity (≤1.001), creatinine (≤50 mg/L), non-physiological salt concentrations, abnormal pH values, absence or abnormally low levels of endogenous steroids, etc. However, beyond these values for specific gravity and creatinine, no concise guidelines have been defined while, at the same time, this particular Technical Document further mandates that the laboratory is obliged to determine the nature of the adulterated sample (examples given below) before an adverse analytical finding can be reported.

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Article highlights

- Cases of sample manipulation in the doping control context have occasionally been identified.
- Various different approaches of urine adulteration/substitution were observed and investigated.
- Underreporting of manipulation/adulteration must be taken into consideration, especially when suspected urine substitution is not pursuable.
- Parameters characterizing a urine sample for routine doping controls are discussed vis-à-vis acceptance criteria of other analytical drug testing disciplines.
- Extending guidance in managing urine substitution or adulteration by regulatory organizations is suggested

A scientific debate might be warranted if that currently enforced requirement of identifying the nature/type of a sample, which is not consistent with human urine, is indeed relevant if the incompatibility of determined markers with human urine has been demonstrated. In that regard, it stands to reason to assess which routine analytical data acquired in the context of mandatory sports drug testing methods can contribute to determine whether a sample is indeed a urine sample. Or, alternatively, considering to adopt (selected) criteria from those suggested by other analytical sciences disciplines in support of assessing a test sample's authenticity.

2. Sample validity test parameters – case reports from routine doping controls

As indicated above, various guidelines and studies have been published that thematize the relevance of robust parameters that testing laboratories are obliged to (or at least recommended to) determine in order to differentiate valid from invalid specimens [16–19]. An overview regarding a set of standard parameters, acceptance criteria, and additional biomarkers contributing to the assessment of a sample's validity is presented in Tables 1 and 2.

Routine doping control analyses have resulted in several findings over the past three decades, where urine substitution or (the attempt of) manipulation of some kind was suspected and/or proven. In some of these cases, the consideration of the parameters summarized in Table 1 would presumably have flagged the provided specimen as invalid; in others, however, the substitution of urine might still have gone undetected.

3. Substitution of urine with another urine or urine manipulation

3.1. Case 1 – urine substitution (1992) [7]

In early 1992, triggered by a first round of suspicious test results concerning the delivery of tampered urine samples, a group of athletes was target-tested at their out-of-competition training camp, and routine screening analyses including (amongst others) steroid profiling were conducted. While standard parameters such as pH and specific gravity were within normal reference ranges, substantial similarities (including the pH and specific gravity values, urine coloration, and consistency and rate (velocity) of sedimentation) were observed, which were further substantiated by the measured steroid profile data. These included at the time the concentrations of androsterone (A), etiocholanolone (Etio), epitestosterone (E), testosterone (T), 11-OH-androsterone (11-OH-A), 11-OH-etiocholanolone (11-OH-Etio), 5α-androstane-3α,17β-diol (5αAdiol), 5β-androstane-3α,17β-diol (5βAdiol), pregnanediol (PD), tetrahydrocortisol (TC), and selected ratios thereof (A/Etio, T/E, 11-OH-A/11-OH-E, and 5αAdiol/5βAdiol). With a total of 16 parameters that were found to be identical under consideration of applied measurement uncertainties, the probability that three samples of different origins yielding identical values for all of these 16 parameters was calculated to be $1{:}3 \times 10^{57}$. Still, an effort was made to provide "fingerprints" of the samples by liquid chromatography-ultraviolet detector (LC-UV) and gas chromatography-nitrogen/phosphorus detector (GC-NPD)

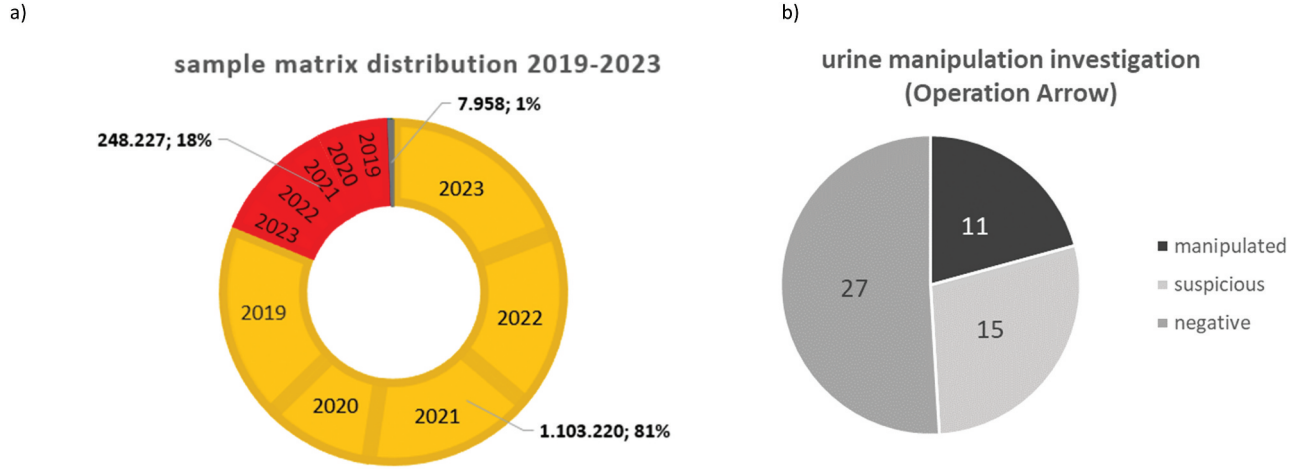

Figure 1. a) sample matrix distribution of doping control samples collected between 2019 and 2023: 1.103.220 urine samples (yellow), 248.227 blood samples (red, incl. serum and whole blood/athlete biological passport specimens), and 7.958 dried blood spot samples (DBS, gray); b) outcome of data and sample reanalysis within the "Operation Arrow" [8], where a total of 53 samples (related to 39 athletes) were included: 11 samples were proven to be substituted, 15 were "very likely" substituted (but already discarded at the time of the investigation), and 27 were found to be not substituted.

Table 1. Selected markers and guidelines for sample validity tests (SVTs) and target analytes indicating sample substitution or manipulation. Sources of information are attributable to areas of a) health and human services, b) workplace drug testing, c) forensic sciences/academia.

	Guideline/Study				
Parameter	Bush 2008 [8][a]	SAMHSA 2017 [24][a]	EWDTS 2017 [11][b]	Goggin etal. 2017 [12][b] (refers to Bush 2008 for standard parameters)	Kyle and Kaur 2020 [14][c] (refers to SAMHSA 2017 for standard parameters)
Standard parameter			*Acceptance range*		
creatinine	≥200 mg/L (20–200 mg/L to be reported as "dilute"; below is considered "substituted" if SG is < 1.0010 or > 1.0200)	≥200 mg/L	≥226 mg/L (56–226 mg/L to be reported as "dilute"; < 56 mg/L to be reported as "specimen validity failed")	≥200 mg/L (20–200 mg/L to be reported as "dilute"; below is considered "substituted" if SG is < 1.0010 or > 1.0200)	≥200 mg/L
pH	< 3 or ≥ 11 is considered adulterated	4.5 – 9 (< 4 or ≥ 11 is considered adulterated)	4 – 9 (< 4 or > 9 is considered adulterated, to be reported as "specimen validity failed")	< 3 or ≥ 11 is considered adulterated	4.5 – 9 (< 4 or ≥ 11 is considered adulterated)
Specific gravity	1.0010 - 1.0200 (determined only in combination with creatinine)	1.0030 – 1.0200 (to be determined when creatinine < 200 mg/L)	1.001 – 1.020 (to be determined when creatinine < 226 mg/L)	1.0010 - 1.0200 (determined only in combination with creatinine)	1.0030 – 1.0200 (to be determined when creatinine < 200 mg/L)
≥1 oxidant (e.g., nitrite, halogens, chromium IV)	Nitrite < 500 µg/mL Chromium IV < 50 µg/mL	Nitrite < 200 µg/mL Chromium IV < 50 µg/mL	Nitrite < 500 µg/mL	Nitrite < 500 µg/mL Chromium IV < 50 µg/mL	Nitrite < 200 µg/mL Chromium IV < 50 µg/mL

analyses. The obtained data further affirmed the specimens' identical composition, corroborated by caffeine and chloroquine (an anti-malaria drug) as well as desethylchloroquine concentrations. In retrospect, the approach employed in 1992 has already been in agreement with the more recently suggested recommendations of including additional biomarkers into the verification process of a urine sample's validity; however, only the comparison of test results of the arguably different urine specimens suggested manipulation in this case. Each individual urine sample would, in all likelihood, have passed the SVT.0

3.2. Case 2 – urine dilution with alcoholic beverage (1998) [3]

An unusually intense odor of a routine doping control urine sample was one of several unexpected parameters determined in 1998. When the A-sample was first opened, a strong (whiskey-like) alcoholic smell was recognized, and while the dipstick specific gravity test did not indicate any anomality, gravimetry measurements yielded a value of 0.98 g/mL. Headspace GC analysis confirmed an ethanol content of 21%, which was considered incompatible with human life and, consequently, sample manipulation was suspected. As further steroid profile analyses corroborated the assumption of testosterone doping with concentrations of A, Etio, 5αAdiol, and 5βAdiol (far) beyond reference ranges known for female athletes as well as carbon isotope ratio analyses supporting the exogenous origin of the target analytes, manipulation with the goal of masking the use of a prohibited substance was established.

3.3. Case 3 – urine manipulation with protease (2009) [4]

Specifically, evading the detection of the illicit use of peptide hormones was, presumably, the reason why urine samples collected in the context of routine doping controls were found to be enriched with the protease bacillolysin in 2009[3] Effectively, all standard parameters such as pH, specific gravity, and lower molecular mass analytes remain unaffected, but relevant target analytes such as erythropoietin receptor agonists, human chorionic gonadotrophin, long-acting growth hormone derivatives, etc., would be rapidly degraded and, consequently, undetectable in routine doping control analyses. Hence, monitoring markers indicative for the presence of exogenous proteases has received growing attention, as this form of manipulation is hardly covered by conventional sample validity test parameters.

4. Substitution of urine with other liquids

4.1. Case 4 – beer [20]

Human urine contains a distinct pattern of naturally occurring and endogenously produced steroidal compounds, including, e.g., T, E, A, and Etio, which yield an individual's "urinary signature" utilized in sports drug testing. If these analytes are not detected in a doping control sample, manipulation is principally proven. However, in order to further corroborate the finding, the identification of the sample's surrogate is frequently requested, and a variety of analytical strategies can be applied. In the case of a urine sample substitution using alcohol-free beer, the specimen was characterized by proteomics methods flanked with GC-MS and GC-FID measurements. Eventually, the identification of

Table 2. Selected complementary markers for sample validity tests (SVTs) and target analytes indicating sample substitution or manipulation. Sources of information are attributable to areas of a) health and human services, b) workplace drug testing, c) forensic sciences/academia.

	Guideline/Study			
	SAMHSA 2017 [24][a]	Goggin etal. 2017 [12][b]	Kluge etal. 2018 [13][c]	Kyle and Kaur 2020 [14][c]
Additional parameter	If the concentration of a biomarker is not consistent with that established for human urine, a sample is reported as "invalid"		Detection rate of < 6 of below named 10 endogenous biomarkers suggests urine adulteration	
Endogenous biomarker				
Normetanephrine		Presence supports validity		
Uric acid		Presence supports validity		
3-methylhistidine		Presence supports validity		
Theobromine		Presence supports validity		Presence supports validity
Theophylline		Presence supports validity		
Urobilin		Presence supports validity		
Caffeine				Presence supports validity
Cotinine				Presence supports validity
Phenylalanine			Presence supports validity	
Tryptophan			Presence supports validity	
Propionyl-carnitine			Presence supports validity	
Butyryl-carnitine			Presence supports validity	
Isovaleryl-carnitine			Presence supports validity	
Hexanoyl-carnitine			Presence supports validity	
Heptanoyl-carnitine			Presence supports validity	
Octanoyl-carnitine			Presence supports validity	
Phenylacetylglutamine			Presence supports validity	
Indoleacetylglutamine			Presence supports validity	
Biological elements				Presence of e.g., leukocytes, erythrocytes, squamous epithelium shown by microscopy supports validity
Marker for synthetic sample origin				
Benzoisothiazolinone		Presence suggests substitution		
Triethylene glycol		Presence suggests substitution		
Tetraethylene glycol		Presence suggests substitution		
Tetrapropylene glycol			Presence suggests substitution	
Pentapropylene glycol			Presence suggests substitution	
Hexapropylene glycol			Presence suggests substitution	
Heptapropylene glycol			presence suggests substitution	
Octapropylene glycol			Presence suggests substitution	
Nonapropylene glycol			Presence suggests substitution	
Decapropylene glycol			Presence suggests substitution	
Undecapropylene glycol			Presence suggests substitution	
Synthetic debris				Presence shown by microscopy suggests urine substitution

barley specific proteins such as serpin-Z4, the presence of hordenine, and an ethanol content of 120 mg/kg (0.015%) supported the assumption that a brewing product rather than a urine sample was provided.

4.2. Cases 5 & 6 – water and saliva-enriched water [21]

A similar case (but literally of different flavor) occurred when a water sample was declared as an athlete's doping control specimen. Again, the absence of any commonly observed human urinary steroid hormone was indicative for sample manipulation, but in contrast to the aforementioned beer, it was the lack of any organic matter (including, e.g., creatinine, urea, uric acid) and the specific gravity of 1.001 that led to the (preliminary) conclusion that not human urine but only water was in the doping control sample container. The comparison to a urine sample collected from the same athlete at a later occasion corroborated the suspicion of manipulation as the analysis of the additional sample yielded levels within commonly observed reference ranges for all tested parameters (Table 3).

In an almost identical case, an athlete's routine doping control sample (referred to here as sample #1) was found to be indistinguishable from the purely aqueous negative quality control specimen employed for steroid profile analyses. While the pH was within acceptance range (7.7), the specific gravity was 1.000, the sample did not exhibit any noticeable odor, and

Table 3. Summary of markers determined in examples of presumed and/or proven sample manipulation cases in routine doping controls.

	Test results (initial testing procedure)							
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
Presumed/proven manipulation	Urine substituted by human urine	Urine diluted with whisky	Urine enriched with protease	Urine substituted by beer	Urine substituted by water	Urine substituted by water (containing saliva)	Urine substituted by tea	Urine substituted by synthetic urine
Markers								
pH	Normal	No information	Normal	Suspicious	Normal	Normal	Normal	Normal
Specific gravity	Normal	Normal (dipstick test) Suspicious (gravimetry)	Normal	Normal	Suspicious	Suspicious	Suspicious	Normal
Color	Normal	Normal	Normal	Normal	Suspicious	Suspicious	Normal	Normal
Odor	Normal	Suspicious	Normal	Suspicious	Suspicious	Suspicious	No information	Not tested
Complementary markers/methods				**Test results**				
Creatinine	Not tested	Not tested	Not tested	Not tested	Suspicious	Suspicious	Suspicious	Not tested
Urea	Not tested	Not tested	Not tested	Not tested	Suspicious	Suspicious	Suspicious	Not tested
Uric acid	Not tested	Not tested	Not tested	Not tested	Suspicious	Suspicious	Not tested	Not tested
Chromatographic-mass spectrometric methods (general)	Suspicious	Suspicious	Normal	Suspicious	Not tested	Not tested	Not tested	Not tested
Steroid profile analysis	Suspicious	Suspicious	Normal	Suspicious	Suspicious	Suspicious	Suspicious	Suspicious
Isotope ratio analysis	Not tested	Suspicious	Not tested	Not tested	Not tested	Suspicious	Not tested	Suspicious
Ggel electrophoresis/ bottom-up proteomics	Not tested	Not tested	Suspicious	Suspicious	Not tested	Suspicious	Not tested	Suspicious
DNA analysis	Not tested	Not tested	Not tested	Not tested	Not tested	Normal	Not tested	Not tested

appeared entirely colorless. Also here, a comparator sample collected from the same individual ca. 3 months later (sample #2) yielded test results within typical reference ranges for all determined parameters. Upon questioning the authenticity of sample #1 as urine, a DNA test to be conducted on the sample was inquired, and the (exclusive) presence of the athlete's DNA was confirmed. However, further analyses, including gel electrophoresis with bottom-up proteomics as well as carbon, hydrogen, oxygen and nitrogen isotope ratio mass spectrometry, revealed a) the presence of salivary α-amylase in the absence of any other commonly observed urinary protein in sample #1, and b) significantly deviating isotope ratio mass spectrometric results for the two samples assigned to the athlete. Values of $\delta^{13}C = -14.13$ ‰ (sample #1) and $\delta^{13}C = -22.01$ ‰ (sample #2) suggest a significantly higher proportion of inorganic carbon (whilst absolute amounts were low) in sample #1 than in sample #2. In addition, $\delta^{2}H = -45.0$ ‰ and $\delta^{18}O = -6.76$ ‰ (both from #1) deviated significantly from #2 ($\delta^{2}H = 6.1$ ‰ and $\delta^{18}O = 0.65$ ‰). Also, suspiciously low/undetectable levels of creatinine, urea, and uric acid were recorded. In combination, all results were compatible with sample #1 being a (tap) water sample with residues of the athlete's saliva and, hence, not urine.

4.3. Case 7 – tea [22]

Also here, specific gravity (1.000) and the lack of creatinine, urea, and any commonly observed human urinary steroid hormone was indicative for sample manipulation, while only the pH (7.4) and color were inconspicuous. Noteworthy was further the peculiar appearance of the frozen sample, which exhibited a darker-colored center and a clear periphery, resembling a rather atypical picture that is not commonly observed with frozen doping control urine samples. The specimen was found to contain caffeine at a concentration of ca. 20 µg/mL, and this overall scenario (i.e., freezing pattern, coloration, and caffeine content) was reconstructed by preparing 200 mL of black tea (2 g of leaves) and diluting the brew 1/10 (v/v) with water. These results corroborated the assumption of tea having been provided instead of a urine sample, constituting an attempt of sample substitution.

4.4. Case 8 – synthetic urine [23]

A doping control urine sample, flagged as suspicious due to the absence of detectable levels of endogenous steroids, was presented with a specific gravity of 1.014 and a pH of 8.3. The presence of urea was assessed using the xanthydrol-method, which further allowed to subsequently analyze the nitrogen isotope ratio of urea (if found). Also, the isotope ratio of the total carbon content of a dried sample aliquot was determined.

The nitrogen and carbon isotopic composition of the suspicious (and allegedly human urine) sample was $\delta^{15}N = -2.0$ ‰ for urea and $\delta^{13}C = -36.5$ ‰ for total carbon, indicating a synthetic origin, comparable to commercially available urea standards and synthetic urine matrices analyzed under identical conditions.

Moreover, when investigated by gel electrophoresis, the pattern of proteins observed following Coomassie-staining was found to be atypical, with merely one intense band at an apparent molecular weight of ~70 kDa plus very few faint additional signals being detected. In-gel tryptic digestion, LC-HRMS/MS analysis and protein database search revealed the presence of bovine albumin (Uniprot accession # P02769) instead of its human analog, clearly demonstrating a case of urine substitution.

5. When is a sample a urine sample?

As summarized in Table 3, manipulated doping control urine samples might pass initial testing procedures including pH, specific gravity, as well as observations concerning color and odor, particularly when a urine specimen was substituted by another human or synthetic urine.

The presumably most robust parameter routinely monitored in doping controls that allows for identifying sample manipulation via the presented methods appears to be the urinary steroid profile. Except for case 3, where adulteration by means of protease addition was uncovered only by complementary gel electrophoretic/mass spectrometric means, either the identity of profiles (where no identity should exist), the presence of atypical urinary steroid concentrations and/or ratios thereof, or the indifference of the alleged urine sample and the purely aqueous negative quality control sample (corresponding to the absence of a steroid profile) reliably provided indication of urine manipulation. In order to account for protease (or glyosylase/glycanase) spiking of doping control urine samples, (glycol)peptidic biomarkers would be a valuable addition for comprehensive analyses concerning adulteration attempts. To date, such attempts of forced analyte degradation are uncovered only using dedicated screening methods, where commonly stable isotope-labeled internal standards matching target peptide sequences are employed. These are degraded in the process and, hence, indicate an atypical urine sample composition, triggering follow-up proceedings.

Follow-up investigations corroborated the suspicion of urine sample manipulation in all instances. Yet, for sports drug testing purposes and, specifically, decision-making processes including result reporting by anti-doping laboratories, the question when a sample is not to be considered as "urine sample" has not been conclusively addressed. Guidelines similar to those applicable in workplace drug testing, toxicology, etc., that classify urine samples as, e.g., 'invalid' could therefore be a valuable addition to anti-doping laboratories. However, here, priority should be given to defining criteria that allow for unequivocally flagging samples that shall not be considered as urine by the anti-doping laboratory (as generally mentioned in the Technical Document EAAS2021 [15]). In such cases, i.e., the reception of a specimen that (contrary to its description) is by definition not a urine sample but, for instance, synthetic urine or water, an adverse analytical finding would be documented.

Such a process would require a defined set of criteria that a doping control urine sample must meet. In consideration of routine doping control proceedings and the thereby

generated analytical data, focusing on pH, steroid profile parameters, the abundant urinary analyte 5-aminoimidazole-4-carboxamide ribonucleoside (AICAR), and specific gravity appears to be a particularly practical approach. Here, aligning with selected aspects of the European Workplace Drug Testing Society (EWDTS) [11] and/or Substance Abuse and Mental Health Services Administration (SAMHSA) [24] guidelines for federal workplace drug testing programs and exploiting analytical data generated at globally operating WADA-accredited anti-doping laboratories, could be a viable option, with e.g.

5.1. Criteria defining a "urine sample"

5.1.1. First-tier parameters (always monitored)

(1) pH (4–11)
(2) presence of a steroid profile (presence of human urinary steroids, as monitored for the steroidal module of the athlete biological passport – steroidal module, within established reference ranges [15,25–33])
(3) urinary steroid test results must be distinguishable from negative water QC sample test results [19,20]
(4) presence of urinary 5-aminoimidazole-4-carboxamide ribonucleoside (AICAR) within established reference ranges [34–37]
(5) specific gravity (>1.001) [38,39]

5.1.2. Second-tier parameters (optionally obtained if first-tier steroid profile and AICAR parameters are suspicious)

(1) creatinine (>50 mg/L)
(2) urea (>750 mg/dL)
(3) presence of 6/10 endogenous biomarkers (see Table 2 [13])
(4) carbon, hydrogen, nitrogen and oxygen isotope ratios compatible with human reference ranges [40–42]

6. Future perspective

In conclusion, a protocol that supports decision-making processes as to when a sample is not to be considered a urine sample in sports drug testing would consist of a mandatory initial screening of the first-tier parameters. First-tier criteria 1–5 must be fulfilled. If not, the sample is to be considered as a urine surrogate.

Further investigations into the nature of the surrogate are not obligatory, but second-tier parameter analyses (6–9) could provide additional insights into the composition and potential origin of the provided liquid. These insights would support follow-up investigations into such cases (where required), as urine sample substitution should, considering the strict regulations of the overall urine sample collection procedure in sports drug testing programs, be extremely rare and difficult to accomplish.

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Author contributions

Thomas Piper: investigation, writing – review and editing, formal analysis; **Hans Geyer**: investigation, writing – review and editing, formal analysis; **Frank Huelsemann**: investigation, writing – review and editing, formal analysis; **Ute Mareck**: investigation, writing – review and editing, formal analysis; **Katja Walpurgis**: methodology, writing – review and editing, formal analysis; **Gregor Fusshoeller**: investigation, writing – review and editing, formal analysis; **Andreas Thomas**: conceptualization, writing – review and editing; **Mario Thevis**: writing – original draft, conceptualization, methodology;

Disclosure statement

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